information

MDPI

Article

Secure Multifaceted-RAG: Hybrid Knowledge Retrieval with Security Filtering

Grace Byun [1,*], Shinsun Lee [1,2], Nayoung Choi [1] and Jinho D. Choi [1,*]

[1] Computer Science and Informatics, Atlanta Campus, Emory University, Atlanta, GA 30322, USA; shinsun.lee@emory.edu (S.L.); nayoung.choi@emory.edu (N.C.)
[2] Hyundai Motor Company, Seoul 06797, Republic of Korea
* Correspondence: grace.byun@emory.edu (G.B.); jinho.choi@emory.edu (J.D.C.)

Abstract

Existing Retrieval-Augmented Generation (RAG) systems face challenges in enterprise settings due to limited retrieval scope and data security risks. When relevant internal documents are unavailable, the system struggles to generate accurate and complete responses. Additionally, using closed-source Large Language Models (LLMs) raises concerns about exposing proprietary information. To address these issues, we propose the **Secure Multifaceted-RAG (SecMulti-RAG)** framework, which retrieves not only from internal documents but also from two supplementary sources: pre-generated expert knowledge for anticipated queries and on-demand external LLM-generated knowledge. To mitigate security risks, we adopt a local open-source generator and selectively utilize external LLMs only when prompts are deemed safe by a filtering mechanism. This approach enhances completeness, prevents data leakage, and reduces costs. In our evaluation on a report generation task in the automotive industry, SecMulti-RAG significantly outperforms traditional RAG—achieving 79.3–91.9% win rates across correctness, richness, and helpfulness in LLM-based evaluation and 56.3–70.4% in human evaluation. This highlights SecMulti-RAG as a practical and secure solution for enterprise RAG.

Keywords: Large Language Model (LLM); information retrieval; Retrieval-Augmented Generation (RAG); domain-specific text generation; security; data confidentiality

check for updates

Academic Editors: Marina Bagic Babac and Andrea Giovanni Nuzzolese

Received: 22 August 2025
Revised: 9 September 2025
Accepted: 12 September 2025
Published: 16 September 2025

Citation: Byun, G.; Lee, S.; Choi, N.; Choi, J.D. Secure Multifaceted-RAG: Hybrid Knowledge Retrieval with Security Filtering. *Information* **2025**, *16*, 804. https://doi.org/10.3390/info16090804

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

Retrieval-Augmented Generation (RAG) [1] has become a powerful tool for AI-driven content generation. However, existing RAG frameworks face significant limitations in enterprise applications. Traditional RAG systems rely heavily on internal document retrieval, which can lead to incomplete or inaccurate responses when relevant information is missing. Moreover, leveraging external Large Language Models (LLMs) like GPT [2], Claude [3], or DeepSeek [4] introduces security risks and high operational costs, making them less viable for enterprise deployment.

To address these challenges, we introduce the **Secure Multifaceted-RAG (SecMulti-RAG)** framework, which optimizes information retrieval, security, and cost efficiency. Our approach integrates three distinct sources: (1) a dynamically updated Enterprise Knowledge Base, (2) pre-written expert knowledge for anticipated queries, and (3) on-demand external knowledge, selectively retrieved when the user prompt is safe. For security, we introduce a filtering mechanism that ensures proprietary corporate data is not sent to external models.

Furthermore, instead of relying on powerful closed-source LLMs, we use a local open-source model as the primary generator, selectively invoking external models only when user prompts are non-sensitive.

In this paper, we apply SecMulti-RAG to the Korean automotive industry, particularly for a report generation task. Our fine-tuned filter, retriever, and generation models show strong performance, ensuring the reliability of our approach. On the report generation task, our method outperforms the traditional RAG approach in correctness, richness, and helpfulness, as evaluated by both humans and LLMs. We also present adaptable strategies to meet specific deployment needs, emphasizing its flexibility. We aim to deploy our system in real-world enterprise environments. Our key contributions are

- A multi-source RAG framework combining internal knowledge, pre-written expert knowledge, and external LLMs to enhance response completeness.
- A confidentiality-preserving filter to prevent exposure of sensitive corporate data to external LLMs.
- A cost-efficient approach that leverages high-quality retrieval to compensate for smaller local generation models.

2. Related Work

2.1. Enhancing Retrieval-Augmented Generation

Many efforts have been made to enhance RAG systems [5,6]. Jeong et al. [7] classify user prompts based on complexity to determine the optimal retrieval strategy, making their approach relevant to our filtering mechanism for selecting retrieval sources. Meanwhile, Yu et al. [8] and Wu et al. [9] replace traditional document retrieval with generative models. In particular, Wu et al. [9] propose a multi-source RAG (MSRAG) framework that integrates GPT-3.5 with web-based search, making it closely related to our work. In contrast, our approach retains internal document retrieval while integrating pre-generated expert knowledge and external LLMs.

2.2. Security Risks in LLM

As generative models are widely used, concerns about security and privacy risks continue to grow. Many studies have explored methods for detecting and mitigating the leakage of sensitive information [10–13]. For example, Chong et al. [14] present a prompt sanitization technique that enhances user privacy by identifying and removing sensitive information from user inputs before they are processed by LLM services. Zhou et al. [15] present a privacy-preserving framework, MixPi, to ensure no sensitive corporate data is exposed by cleverly obfuscating queries before calling external LLMs. Our study incorporates a user-prompt-filtering mechanism, ensuring a more secure retrieval process.

Compared to prior works that primarily combine internal retrieval with web search and closed-source LLM augmentation, our design explicitly triangulates three sources with distinct roles: (i) enterprise-internal documents for provenance-grounded facts, (ii) pre-written expert knowledge to mitigate missing-document gaps in enterprise settings, and (iii) on-demand external knowledge gated by a confidentiality filter. This separation allows us to decouple completeness (via expert and external knowledge) from confidentiality (via filtering and a local generator), while retaining conventional internal retrieval.

3. Method

As shown in Figure 1, our framework consists of three core components: multi-source retrieval, a confidentiality-preserving filtering mechanism, and local model adaptation.

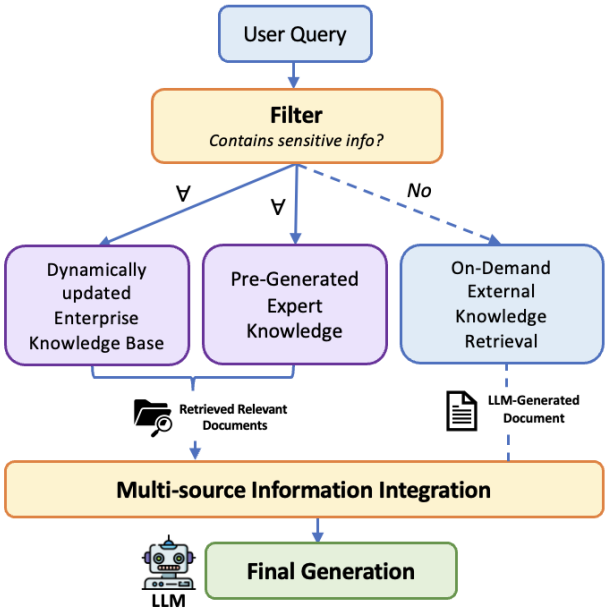


Figure 1. SecMulti-RAG framework. After the filtering prompt is invoked, the system routes the query either to the external-LLM branch (if classified as safe) or to the internal+expert-only branch (if classified as sensitive).

(1) Multi-Source Retrieval

Unlike conventional RAG frameworks that rely solely on internal structured document chunks, our system retrieves information from three distinct sources: (1) internal corporate documents, (2) pre-generated high-quality answers to anticipated queries, and (3) real-time external knowledge generated by closed-source LLMs. This multi-source retrieval strategy improves response completeness and accuracy, especially when internal documents are insufficient.

(2) Confidentiality-Preserving Filtering Mechanism

To mitigate the risk of unintended data leakage when interacting with external closed-source LLMs, we introduce a query filtering mechanism that detects security-sensitive content. If a user query is classified as containing confidential information, external retrieval is skipped, and the system generates responses solely based on internal documents and pre-curated expert knowledge. This mechanism ensures data confidentiality while maintaining retrieval quality (Section 4).

(3) Local Model Adaptation

Since enterprises often encounter security and cost limitations when using powerful closed-source LLMs, we use open-source Qwen-2.5-14B-Instruct [16] as our primary generation model. We fine-tune this model using domain-specific data to better reflect the language and knowledge of the Korean automotive domain (Section 6).

(4) End-to-End RAG Pipeline

The final system consists of a multi-stage pipeline where user queries are processed through filtering, retrieval, and generation. By integrating high-quality retrieval with local model adaptation, our framework shows that a well-optimized retrieval system compensates for the limitations of smaller, locally deployed LLMs, making enterprise RAG both scalable and secure.

4. Confidentiality Filter

4.1. Dataset

Security-sensitive and general (non-sensitive) queries are created by Korean automotive engineers with the assistance of the Claude 3.7 Sonnet [3], accessed through a university-internal service built on AWS Bedrock (https://aws.amazon.com/ko/bedrock/, accessed on 5 January 2025). This setup provides secure access to Claude without exposing data to external LLM providers (all uses of the Claude 3.7 Sonnet model in this study—for data generation and LLM-based evaluation—were accessed exclusively through this secure university-internal service).

The prompts are designed to elicit three types of queries: (1) **general queries** that do not pose confidentiality risks, (2) **security-sensitive queries (easy)** containing explicit project names, and (3) **security-sensitive queries (hard)** that omit project names. Type (3) queries are challenging to classify even for expert engineers due to the absence of clear identifiers. In addition to the queries and the binary labels (sensitive or non-sensitive), brief rationales are generated to explain the reasoning behind each label. Table 1 summarizes data statistics. Figure A1 illustrates the prompt template used to construct the query dataset, and Figure A3 provides the prompt that we use to filter out user queries that should not be exposed to closed-source LLMs.

Table 1. Data split for training and evaluation of the filter model.

Set	General	Unsafe		Total
		Easy	Hard	
Train	820	800	720	2340
Validation	102	100	90	292
Test	103	101	90	294

4.2. Model

For the filter, we fine-tune a lightweight model, Qwen2.5-3B-Instruct [16], to classify safe and unsafe queries. It is fine-tuned for three epochs with a learning rate of 8×10^{-6}, using a batch size of 2 and gradient accumulation of 256 to simulate a large batch. Training is conducted on 3 $\times$ RTX A6000 (48 GB).

4.3. Evaluation

As shown in Table 2, we evaluate the filter model on two subsets: **Easy-only** (queries with explicit project names) and **Easy&Hard** (both with and without project names). Security-sensitive queries (class 0) are treated as the positive class, making recall critical in preventing information leakage to external LLMs. For easy test cases, the filter achieves 99.01% recall, indicating strong performance with minimal false negatives. When ambiguous queries are included, accuracy drops to 82.31% and recall to 74.35%, while precision remains high (97.93%), meaning that when it does flag a query as sensitive, it is highly likely to be correct. To estimate the upper bound, we conduct human evaluation on the **Easy&Hard** set. The expert annotator achieves 80.95% accuracy, 92.99% precision, and 76.44% recall, highlighting the intrinsic ambiguity and difficulty of the task.

Table 2. Evaluation results of the filter model and human annotators on the test set. Easy-onlyincludes security-sensitive queries with explicit project names. Easy&Hardincludes both easy (with project names) and hard (without project names) queries. Humanshows expert-labeled upper bound performance.

Method	Test Data	Acc %	Prec	Rec
Filter	Easy-only	98.04	97.09	99.01
	Easy&Hard	82.31	97.93	74.35
Human	Easy&Hard	80.95	92.99	76.44

4.4. Application

In practical deployment, constructing a labeled dataset of safe and unsafe queries for filter training can be labor-intensive. To address this, we propose a progressive deployment strategy for the confidentiality filter. Initially, the system operates without filtering or external retrieval, relying only on internal documents and pre-generated expert knowledge. During this phase, real user queries are collected and later labeled to train a filter model, enabling cost-effective integration over time. Alternatively, the filter can perform query rewriting—flagged queries are transformed into safer versions, allowing secure forwarding to external LLMs. This flexible design supports scalable adaptation to organizational privacy and deployment needs.

5. Retrieval

This paper focuses on generating reports for enterprise-level engineering problems. Our retrieval system is built upon 6165 chunked documents, consisting of 5625 chunks from the Enterprise Knowledge Base and 540 from Pre-written Expert Knowledge. To prevent data leakage, only the training subset of the 675 Pre-written Expert Knowledge documents is included in the retrieval pool; the remaining 135 keyword–report pairs are reserved for the final evaluation of **SecMulti-RAG** (Section 7). All documents listed in Table 3 are indexed using FAISS [17], and our trained retriever retrieves the top five most relevant documents for each query.

Table 3. Overview of chunked documents used in SecMulti-RAG retrieval. Traditional RAG retrieves only from the Enterprise Knowledge Base. The column "File/Page" indicates the number of files for test reports and meeting reports and the number of pages for the textbook.

Type	Source	File/Page	Chunks
Enterprise Knowledge Base	Test Report	1463	4662
	Meeting Report	249	882
	Textbook	404	81
Pre-written Expert Knowledge	Gold Report	-	540
Total			6165

5.1. Dataset

5.1.1. Enterprise Knowledge Base

For the Enterprise Knowledge Base dataset, we use the dataset introduced by Choi et al. [18]. As shown in Table 4, it consists of test reports, meeting reports, and the textbook Crash Safety of Passenger Vehicles [19], with each document segmented into meaningful units such as slides, chapters, or other relevant sections. QA pairs from the reports and textbook are used to train both the retriever and the generation model.

Table 4. Enterprise Knowledge Base dataset: statistics by source, detailing the distribution of chunks and QA pairs.

Source	Data	Train	Val	Test	Total
Test Report	Chunk	3729	466	467	4662
	QA Pair	47,660	5823	5919	59,402
Meeting Report	Chunk	705	88	89	882
	QA Pair	6144	752	800	7696
Textbook	Chunk	64	8	9	81
	QA Pair	1182	162	161	1505

5.1.2. Pre-Written Expert Knowledge

To construct a high-quality knowledge source, a domain expert in automotive engineering first curates a list of domain-specific keywords, representing expertise-level problems. Using these keywords, the expert then generates pre-written expert knowledge. A total of 675 keyword–report pairs are partitioned into training, validation, and test splits in an 8:1:1 ratio. Below are some examples of the keywords we use to generate pre-written expert knowledge. Reports related to these keywords (main problems) are pre-generated by automotive engineers (see Figure A2 in Appendix A for details):

- A필러 상부 강성 부족 및 변형 (Insufficient upper stiffness and deformation in A-pillar);
- A필러 힌지 크랙 발생 (Hinge crack formation in A-pillar);
- A필러와 대시 연결부 찢어짐 (Tearing at the connection between the A-pillar and the dashboard);
- A필러 변형으로 인한 윈드실드 파손 (Windshield damage due to A-pillar deformation).

5.1.3. External Knowledge from LLM

After the user query passes the safety filter and is deemed safe, we use GPT-4o (https://platform.openai.com/docs/models/gpt-4o, accessed on 5 January 2025) to provide on-demand external knowledge. Specifically, in this paper, it generates a general-purpose technical background document to assist engineers in drafting formal safety reports. The generated document is then indexed into the document pool for future retrieval. The prompt is illustrated in Figure 2.

Prompt for External Knowledge Retrieval

```
You are an expert in automotive crash safety engineering.
You are tasked with generating a general-purpose technical background document **in Korean** that can assist engineers in drafting formal safety
      reports.
Only a high-level keyword describing a common engineering issue is provided.
Based on your expert knowledge, please provide background information and reasonable technical discussion,
without fabricating any specific case data, test results, or numeric values that are not grounded in well-known general principles.

Your output should include the following sections:
1. **Title**: A concise technical title derived from the given keyword.
2. **Issue Overview**:
   - Describe the typical nature of this issue in automotive safety engineering.
   - Mention which vehicle components are generally involved.
   - Explain how this issue may impact structural integrity or passenger safety.
3. **Common Engineering Considerations**:
   - Discuss known design challenges, structural constraints, or typical causes.
   - Do NOT include fabricated test results or data.
   - Only refer to general engineering principles or commonly reported concerns in literature.
4. **Improvement Suggestions (General)**:
   - Suggest generic design or material improvements.
   - Clarify expected benefits and rationale (without specific test results).
5. **Conclusion**:
   - Summarize key considerations.
   - Emphasize that this document provides only general technical guidance, not case-specific findings.

Important:
- Do NOT fabricate test results, measured data, or specific incident cases.
- Only refer to general trends, common engineering knowledge, and design principles.
- This is intended to serve as a general technical reference, not a case analysis.

keyword: {keyword}
```

Figure 2. Prompt used for generating on-demand external knowledge using GPT-4o.

5.2. Retriever

We fine-tune BGE-M3 [20], a multilingual encoder supporting Korean, using QA and keyword–report pairs (Sections 5.1.1 and 5.1.2). Training is conducted for 10 epochs on 4 × RTX A6000 (48 GB) with publicly available code [20–23]. We use dense retrieval only and do not apply a reranker; top-*k* candidates are taken directly from the vector index. For evaluation, we use all splits (training, validation, and test) as the chunk pool to ensure sufficient data coverage and mitigate potential biases due to the small size of the test set.

5.3. Retriever Evaluation

The performance of the retriever is evaluated using Mean Average Precision (MAP@k), which calculates the average precision of relevant results up to rank k. As shown in Table 5, fine-tuning BGE-M3 leads to significant improvements, underscoring the importance of task-specific adaptation.

Table 5. Comparison of retrieval performance between the vanilla and fine-tuned models on our test dataset.

Model	MAP@1	MAP@5	MAP@10
BGE (Vanilla)	0.2855	0.3793	0.3925
BGE (Fine-tuned)	**0.5965**	**0.7027**	**0.7099**

Note. Bold values indicate the best performance in each column.

5.4. Document Selection Strategy

In this study, we rank candidate documents by semantic similarity and apply a selection constraint: at most one external knowledge source is included per query. GPT-generated documents are limited to one per query as they provide only general technical background, and excessive reliance on such external content may reduce the factual grounding of responses. Limiting external LLM-generated background to one document per query reduces (i) context dilution and (ii) hallucination risks, while keeping auditability of non-proprietary content. Although this constraint is currently implemented via heuristic rules, we aim to develop a learning-based document selection strategy that jointly considers query characteristics and document provenance.

6. Generation

6.1. Generator

We use Qwen-2.5-14B-Instruct [16] as our base language model, as it is one of the few multilingual models that officially support Korean while offering a sufficient context length. Full fine-tuning of the 14B model was performed for three epochs using 3 × H100 (80 GB), with a batch size of 2, gradient accumulation of 64, and a learning rate of 2×10^{-5}. We conduct LoRA fine-tuning using QA pairs introduced in Section 5.1.1.

6.2. Result

Table 6 summarizes the retrieved document sources and filtering results for the 135 test queries. All queries retrieved pre-generated expert knowledge, while 28.1% and 18.5% also retrieved GPT-generated and internal documents, respectively. Among the test queries, 34 are classified as safe, enabling on-demand GPT generation. Interestingly, the number of queries retrieving GPT-generated documents slightly exceeds the number of safe queries. This is because previously generated external documents remain in the retrieval pool and can still be retrieved for relevant future queries, even if those queries are classified as sensitive. This illustrates a key benefit of our system: as more external knowledge is

accumulated over time, the retrieval pool becomes richer, allowing even sensitive queries to benefit from external knowledge without compromising security.

Table 6. Distribution of retrieved document types (appearing at least once among the top 5 documents) and filtering outcomes in the SecMulti-RAG.

Category	Count (%)
Pre-written knowledge retrieved	135 (100%)
External knowledge retrieved	38 (28.1%)
Internal document retrieved	25 (18.5%)
Filter = 1 (safe)	34 (25.2%)
Filter = 0 (security-sensitive)	101 (74.8%)

7. Evaluation

7.1. Method

To evaluate the effectiveness of our approach, we conduct a qualitative assessment based on three metrics: correctness, richness, and helpfulness. We perform pairwise comparisons using both LLM-as-a-judge [24] and human evaluation, comparing responses generated by **Traditional RAG**—which retrieves only from the internal knowledge base—with our **SecMulti-RAG**, which retrieves from the internal knowledge base, pre-generated expert knowledge, and on-demand external knowledge. An expert-generated test dataset asks for report generation regarding the expert-level automotive domain. For each metric, Claude and a human annotator assess which response (A or B) is better and record the outcome as a win, loss, or tie. To mitigate position bias from the judge LLM, we anonymize the response order by randomly assigning either SecMulti-RAG or Traditional RAG as response A or B in half of the cases (the prompt that we use for LLM-based evaluation is in Figure A4):

- **Correctness** assesses the factual consistency with the given gold answer. The pre-written reports are provided as gold answers.
- **Richness** evaluates the level of detail and completeness in the response.
- **Helpfulness** measures how clear, informative, and useful the response is.

7.2. Result

Figure 3 presents a comparison of win rates between the two systems, evaluated by human annotators and an LLM-based evaluation. Both evaluation sources consistently prefer the outputs of SecMulti-RAG, particularly in the richness metric.

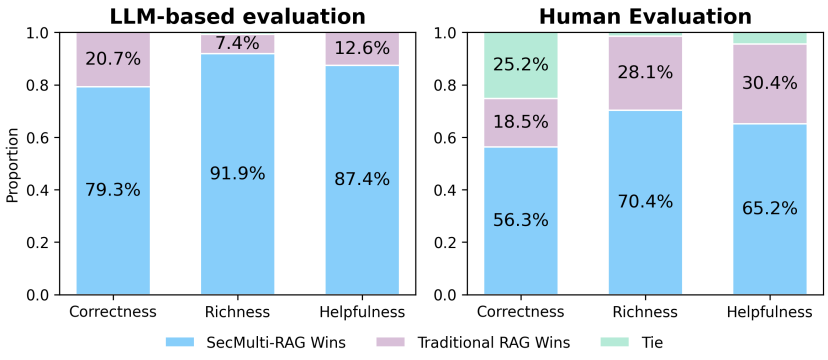


Figure 3. Win rate comparison between SecMulti-RAG and Traditional RAG across evaluation metrics.

7.3. Analysis

While human evaluators tend to assign 'tie' labels more frequently than the judge LLM, both clearly favor SecMulti-RAG across all metrics. Figure 4 illustrates an example of reports generated by SecMulti-RAG and Traditional RAG.

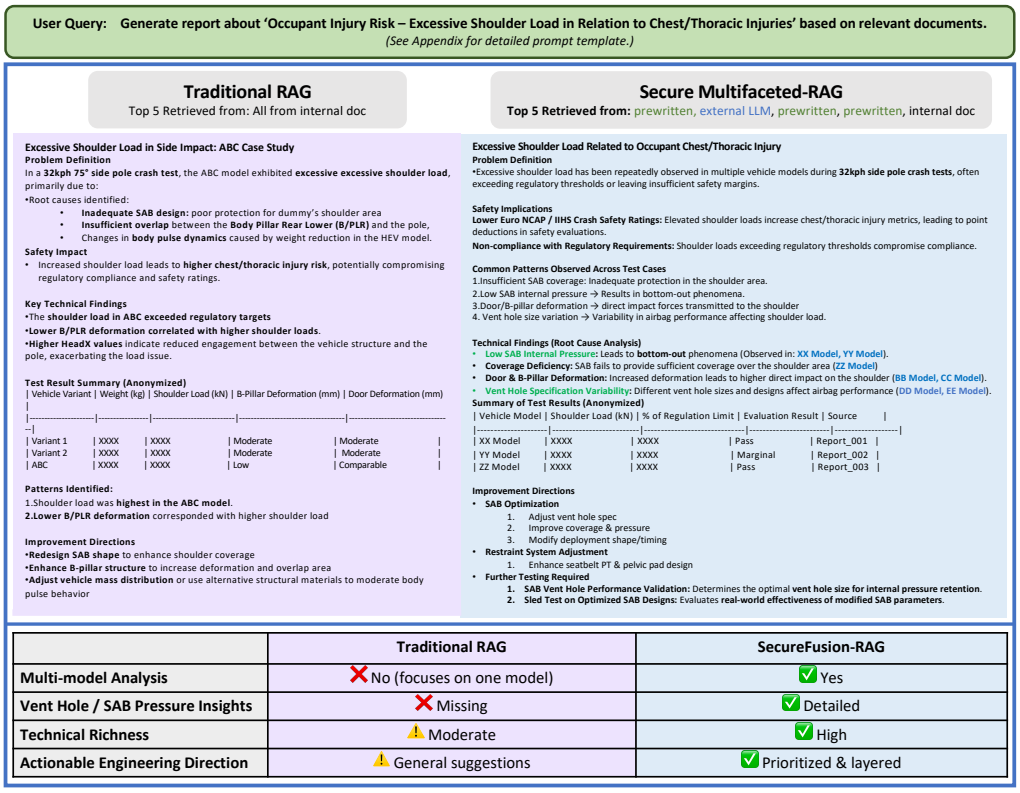


Figure 4. Comparison between Traditional RAG and SecMulti-RAG (translated from Korean; sensitive information anonymized).

As expected, richness is the most notably improved aspect of our framework, reflecting the benefit of incorporating diverse documents from multiple sources. SecMulti-RAG outputs contain significantly more detailed information, such as more diverse test cases and technical findings. In fact, the average length of reports generated by SecMulti-RAG is 2660.21 tokens, compared to 1631.84 tokens from Traditional RAG. In terms of correctness, both systems generally produce factually accurate content but occasionally fail to cite the correct source document. For helpfulness, while our approach delivers more comprehensive reports, Traditional RAG may be more favorable when the engineer's intent is to focus on a specific test case, as its responses tend to be more narrowly scoped. This could potentially be mitigated through prompt tuning. One notable issue is the occasional generation of Chinese characters, due to the Qwen model's Chinese-centric pretraining. This issue is likely caused by longer retrieved documents increasing the context length, which in turn makes the model more prone to such errors.

Table 7 reports the agreement rates and Gwet's AC1 scores. Gwet's AC1 is used instead of Cohen's Kappa due to class imbalance in the evaluation, where SecMulti-RAG is consistently favored over Traditional RAG. The overall agreement between Claude and human judgments is substantial, particularly in richness and helpfulness. The correctness dimension shows relatively lower agreement, largely because the LLM tends to avoid assigning "tie" labels, which human annotators use more frequently. See Figure 5 for confusion matrices.

Table 7. Agreement between Claude and human evaluation results.

Metric	Agreement (%)	Gwet's AC1
Correctness	56.74%	0.4295
Richness	73.05%	0.6812
Helpfulness	69.50%	0.6264
Overall	72.34%	0.6647

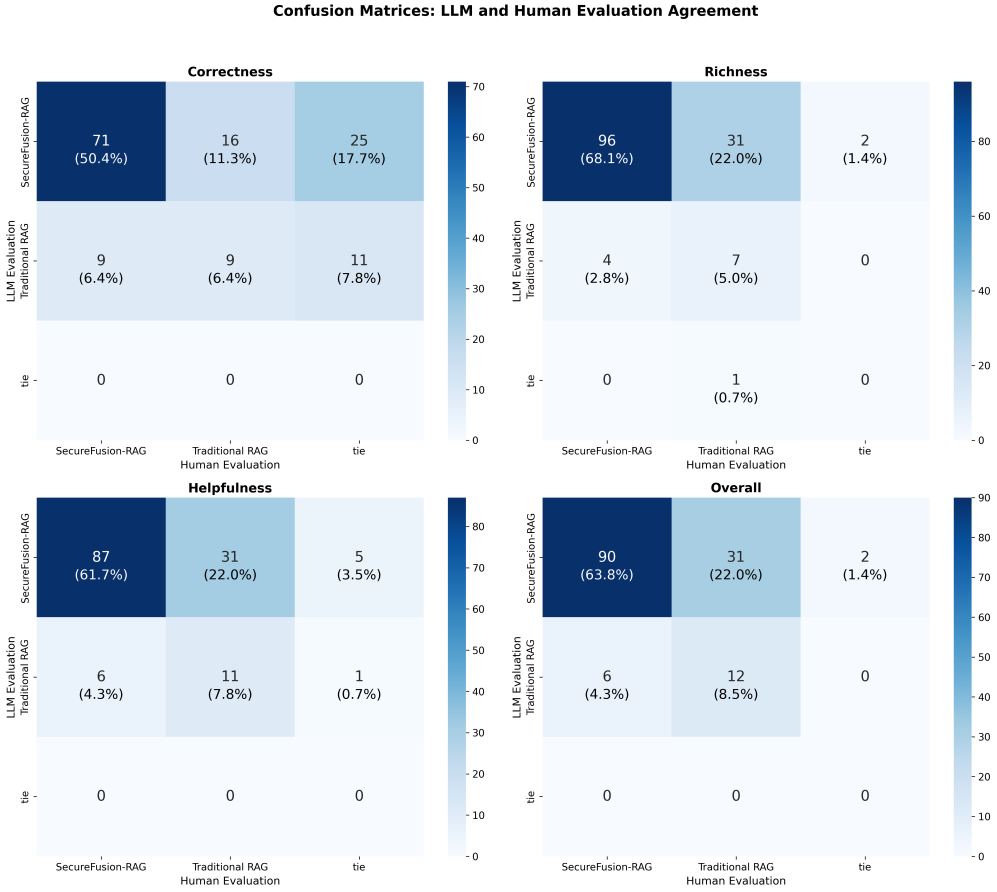


Figure 5. Confusion matrices showing the agreement between LLMs and human evaluations. Most counts lie in the diagonal cells, indicating consistent agreement between both evaluators.

7.4. Application

In this study, we primarily evaluate our RAG framework on the report generation task. However, in practice, the framework is scalable to various tasks and domains. We conduct a preliminary test with a few engineering questions in the automotive domain, as shown in Figure A5 in Appendix B. The SecMulti-RAG responses include specific injury types, structural causes, and implications for official safety assessments, demonstrating greater richness and helpfulness compared to the Traditional RAG responses. This shows that our framework is scalable beyond report generation.

Furthermore, our framework is also cost-effective. For instance, generating lengthy reports with an average of 2660 tokens per query costs only about USD 2–3 in total for 100 queries when using GPT-4o selectively, highlighting the practicality of SecMulti-RAG for enterprise deployment.

8. Limitations

Pre-written expert knowledge can introduce topical bias: issues not anticipated by experts may be underrepresented. Also, there may be increased response latency due to

the additional filtering stage and multi-source retrieval process. The filter trades recall for safety; false negatives pose a leakage risk, while false positives reduce recall of helpful external knowledge. Lastly, due to the lack of publicly available Korean-language datasets in the automotive domain, our evaluation is limited to the report generation task based on a relatively small amount of data that we have constructed ourselves. While this work is intended for an industry track and demonstrates practical significance, future research could enhance the academic impact by showing the scalability of the SecMulti-RAG framework across a broader range of tasks and domains. In fact, we have conducted preliminary experiments on engineering question answering tasks beyond report generation and observed that SecMulti-RAG also performs well in those scenarios, indicating its potential.

9. Conclusions

In this paper, we present SecMulti-RAG for enterprises that integrate internal knowledge bases, pre-generated expert knowledge, and on-demand external knowledge. Our framework introduces a confidentiality-aware filtering mechanism that protects security-sensitive user prompts by bypassing external augmentation when necessary, mitigating the risk of information leakage to closed-source LLMs. In our experiments on automotive engineering report generation, SecMulti-RAG showed clear improvements over Traditional RAG in terms of correctness, richness, and helpfulness. It achieved win rates ranging from 56.3% to 70.4%, as evaluated by human evaluators, outperforming traditional RAG across all metrics. Beyond performance gains, our approach is a cost-efficient, privacy-preserving, and scalable solution, leveraging high-quality retrieval with locally hosted LLMs.

Author Contributions: Conceptualization, G.B. and J.D.C.; Data curation, S.L.; Formal analysis, G.B.; Funding acquisition, J.D.C.; Investigation, G.B.; Methodology, G.B. and J.D.C.; Project administration, S.L. and J.D.C.; Resources, S.L.; Software, G.B. and N.C.; Supervision, J.D.C.; Validation, S.L. and N.C.; Visualization, G.B.; Writing—original draft, G.B.; Writing—review and editing, G.B., S.L., N.C. and J.D.C. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data are not publicly available due to confidentiality agreements with Hyundai Motor Company.

Acknowledgments: We gratefully acknowledge the support of the Hyundai Motor Company. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of Hyundai Motor Company.

Conflicts of Interest: Shinsun Lee was employed by "Hyundai Motor Company". The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

LLM	Large Language Model
RAG	Retrieval-Augmented Generation
BGE	BAAI (Beijing Academy of Artificial Intelligence) General Embedding
SecMulti-RAG	Secure Multifaceted-RAG

Appendix A. Prompts

Figure A1 shows the prompt for query data generation.

To generate pre-written expert knowledge, Korean automotive engineers employ the prompt shown in Figure A2, refined through extensive prompt tuning. The generated reports include a structured composition of problem definition, technical analysis, case analysis, improvement suggestions, and relevant references, and they are subsequently reviewed by domain experts. The keywords provided in Section 5.1.2 serve as the core problem topics addressed in each report. University-internal Claude service from AWS Bedrock is used for security.

Figure A3 presents the prompt for classifying user queries by the presence of sensitive information, and Figure A4 depicts the prompt for LLM-based evaluation.

Prompt

```
[Security-sensitive queries without project name]

You are an engineer in charge of vehicle crash safety testing. Based on internal reports, you are tasked with
      generating sensitive security-related queries that should not be exposed to the public. The queries must
      meet the following requirements:

Query Generation Guidelines:
- Do NOT include any specific project names or test IDs.
- Do NOT ask about exact internal figures or detailed designs directly. Instead, formulate questions that
      indirectly imply such information through the context.
- Each question should be capable of revealing internal issues or technical weaknesses if exposed externally.
- Vary the length of the queries (mostly short questions, but some longer ones included).
- Generate 50 questions in total.
- Provide a one-line explanation for why each question is considered security-sensitive.

Example Questions:
- "What is the root cause of the excessive B-pillar deformation observed in recent tests?" (Indirectly reveals an
      internal issue)
- "What causes the delayed airbag deployment timing?" (Hints at internal system problems)
- "Why is reinforcement needed in the door hinge area?" (Implies structural vulnerability)
- "Why does a specific area repeatedly exhibit excessive deformation in crash tests?" (Reveals unresolved
      structural weaknesses)

Please refer to the examples above and generate 50 diverse security-sensitive questions that indirectly reveal
      sensitive internal information.

Here are the internal reports:
<Reports>
{reports}
</Reports>

================================================================================================================
[Security-sensitive queries with project name]

You are an engineer in charge of vehicle crash safety testing. Based on internal reports, you are tasked with
      generating security-related queries that include a specific project name but exclude test IDs. The queries
      must meet the following requirements:

Query Generation Guidelines:
1. MUST include a specific project name (but do NOT include test IDs).
2. In most queries, place the project name at the beginning of the sentence, but vary the position in some
      queries (middle or end).
3. It is acceptable to include sensitive internal figures or specific design information.
4. The queries must be security-sensitive and unsuitable for public exposure.
5. Vary the length of the queries (mostly short, but some longer ones included).
6. Provide a one-line explanation for why each question is considered security-sensitive.
7. Maintain the following consistent output format:

- "Query sentence"
(Explanation)

Here are the internal reports:
<Reports>
{reports}
</Reports>


================================================================================================================
[General (non-sensitive) queries]

You are an engineer in charge of vehicle crash safety testing. Based on internal reports, generate general
      queries that do not pose any security concerns and can be disclosed publicly. These queries should reflect
      realistic engineering questions that professionals may encounter in practice. The queries must meet the
      following requirements:

Query Generation Guidelines:
1. Do NOT include any specific project names or test IDs.
2. Avoid questions about sensitive internal figures or detailed designs. Instead, focus on general technical
      concepts, practical engineering concerns, and issues that engineers would typically discuss.
3. Questions should be centered on issues arising in crash safety testing environments, engineering decision-
      making, and practical field concerns.
4. Vary the length of the queries (mostly short, but some longer ones included).
5. Provide a one-line explanation for why each question is NOT considered security-sensitive.
6. Maintain the following consistent output format:

- "Query sentence"
(Explanation)

Here are the internal reports:
<Reports>
{reports}
</Reports>
```

Figure A1. Prompt used for query data generation (translated to English).

Prompt

```
# You are an expert in writing professional engineering reports in the field of automotive crash safety. Based on
       the given problem situation and related report data, you must write a technical report that is useful for
      field engineers.

## Input Format
1. **Problem Situation**: <Problem> {problem} </Problem>
2. **Reference Reports**: <Reports> {reports} </Reports>

## Output Requirements
1. **Title**: A technical title that clearly reflects the problem situation.
2. **Problem Definition**:
   * Detailed description of the problem occurrence.
   * Specification of affected vehicle components.
   * Analysis of the impact on safety.
   * Identification of common problem patterns observed across multiple cases.
3. **Technical Analysis**:
   * Root cause analysis (based on report evidence).
   * Summary of related test results and pattern identification.
   * Inclusion of measured numerical data (if available).
   * Comparative analysis and correlation between multiple tests.
4. **Case Comparison Analysis**:
   * List and categorize all cases where similar issues occurred.
   * Compare and analyze similarities and differences between cases.
   * Compare case results in tabular format (recommended).
5. **Improvement Directions**:
   * Summarize all improvement measures mentioned in the reports.
   * Clearly explain the reasons and expected effects of each improvement.
   * Provide a prioritization of improvement measures.
   * Identify areas requiring additional testing and justify their necessity.
6. **Conclusion**:
   * Summary of key issues.
   * Summary of major findings.
   * Comprehensive expected effects of the proposed improvements.
7. **References**:
   * All information sources must be cited in parentheses right after the corresponding content (e.g., "Due to
         insufficient upper A-pillar stiffness, the DASH deformation exceeded the target of 100mm by reaching 128
         mm (C200728A_CV).")
   * Arrange citations appropriately to avoid disrupting readability.

## Key Guidelines
1. **Pattern Identification**: Actively identify and analyze recurring problem patterns across different tests
     and situations.
2. **Data Integration**: Integrate similar data from multiple reports to provide a more comprehensive analysis.
3. **Importance Evaluation**: Assign higher importance to repeated problems and explicitly indicate this.
4. **Flexible Report Structure**: Use the given structure as a starting point but modify, merge, or add sections
     as needed based on data and analysis results.
5. Exclude unnecessary introductions or background explanations and focus on core technical content.
6. Do not make assumptions or use general knowledge to fill in missing report data.
7. Provide specific technical information that engineers can use immediately.
8. Ensure all information can be traced back to its source, but citations should not overshadow the main content.
9. Include relevant data, figures, and specific specifications whenever possible.
10. Focus on the methodology applied to solve the problem and the results achieved.
11. Citations should be concise, placed at the end of the sentence or paragraph, and include only the report code
      in parentheses.

## Citation Examples
* Incorrect: "According to report C200728A_CV, due to insufficient upper A-pillar stiffness..."
* Correct: "Due to insufficient upper A-pillar stiffness, the DASH deformation exceeded the target of 100mm by
      reaching 128mm (C200728A_CV)."
* Multiple citations: "BrIC is an index developed to assess the risk of brain injury caused by head rotational
      motion (C200728A_CV, C200730B_NE)."
* Similar issue citation: "The A-pillar deformation issue was consistently observed in three tests (C200728A_CV,
      C200805B_CV, C200901A_CV), and in all cases, the deformation exceeded the allowable limit by 20-25%."

## Basic Report Structure
‘‘‘markdown
# Technical Title Based on the Problem Situation
## Problem Definition
- Detailed description of the problem and pattern identification.
- Affected vehicle components.
- Safety impact analysis.
## Technical Analysis
- Root cause analysis.
- Summary of test results.
- Measurement data.
## Case Comparison Analysis
| Test ID   | Issue Description | Measured Value | Standard Value | Deviation | Notes  |
|-----------|-------------------|----------------|----------------|-----------|----------------|
| TEST-001  | Issue description | 00.0           | 00.0           | 00.0      | Additional info |
| TEST-002  | Issue description | 00.0           | 00.0           | 00.0      | Additional info |
## Improvement Measures and Effects
- Applied solutions.
- Effectiveness measurement results.
- Technical specifications.
## Improvement Directions
- **Priority 1**: [Improvement measure] - [Clear reason and expected effect]
- **Priority 2**: [Improvement measure] - [Clear reason and expected effect]
- **Additional Tests Needed**: [Test details] - [Reason for necessity]
## Conclusion
- Summary of key issues.
- Summary of major findings.
- Comprehensive expected effects of improvements.
```

Figure A2. Prompt used for generating gold reports. The prompt is originally Korean.

Prompt for Filtering Process

```
You are a classification model for user queries.
There are **TWO CLASSES ONLY**:
0 => The query may contain internal or sensitive information and is NOT safe.
1 => The query is general and can be safely processed.

**IMPORTANT INSTRUCTION**:
- If the query contains any internal project names, testing scores, engineering design details,
weakness (defects), and internal solution, you MUST classify as 0.
- If uncertain, choose 0.

You must output ONLY one number: either 0 or 1.

Examples:

    [Example 1] (GPT Not Allowed - Corporate Internal Information)
    User Question: "Tell me the test results of LX2 P2."
    Judgment: 0

    [Example 2] (GPT Not Allowed - Requesting Internal Evaluation Report for a Specific Vehicle)
    User Question: "Share the internal crash test report of the G70 frontal crash test."
    Judgment: 0

    [Example 3] (GPT Allowed - General Automotive Safety Regulations)
    User Question: "Explain the U.S. Federal Motor Vehicle Safety Standard."
    Judgment: 1

    [Example 4] (GPT Allowed - Structural Improvement Cases for Automotive Components)
    User Question: "Tell me about cases of subframe failure and possible improvements."
    Judgment: 1

    [Example 5] (GPT Not Allowed - Confidential Design Information of a Specific Company)
    User Question: "Tell me the key design changes in Hyundai's new engine blueprint."
    Judgment: 0

    [Example 6] (GPT Allowed - General Mechanical Engineering Theory)
    User Question: "What are some methods to increase the rigidity of a car chassis?"
    Judgment: 1

    [Example 7] (GPT Not Allowed - Requesting Internal Test Data)
    User Question: "Tell me the results of the in-house crash tests conducted by HMG."
    Judgment: 0

    [Example 8] (GPT Allowed - Public Data-Based Information)
    User Question: "Explain the European NCAP crash test standards and evaluation criteria."
    Judgment: 1
```

Figure A3. Prompt used for filtering progress (translated to English).

LLM-evaluation Prompt

```
You are a strict evaluator of AI-generated text for automobile
engineering questions.
You must judge two responses, A and B, on the following metrics:
    (1) Correctness: factual correctness based on the question
    (2) Richness: level of detail and completeness
    (3) Helpfulness: clarity, directness, and overall usefulness

Then decide who wins each metric (A/B/tie), and also decide the
final overall winner.
Output format strictly:
    CorrectnessWINNER: [A/B/tie]
    RichnessWINNER: [A/B/tie]
    HelpfulnessWINNER: [A/B/tie]
    OVERALLWINNER: [A/B/tie]
No explanations.

Question: {question}
Response A: {answer_a}
Response B: {answer_b}
Please provide the winners in the specified format WITHOUT any
numerical scores.
```

Figure A4. Prompt used for LLM-based evaluation.

Appendix B. Framework Application

In this study, we primarily evaluate our RAG framework on the report generation task. However, in practice, the framework is scalable to various tasks and domains. We conducted a preliminary test with a few engineering questions in the automotive domain, as shown in Figure A5. The SecMulti-RAG responses include specific injury types, structural causes, and implications for official safety assessments, demonstrating greater richness and helpfulness compared to the Traditional RAG responses.

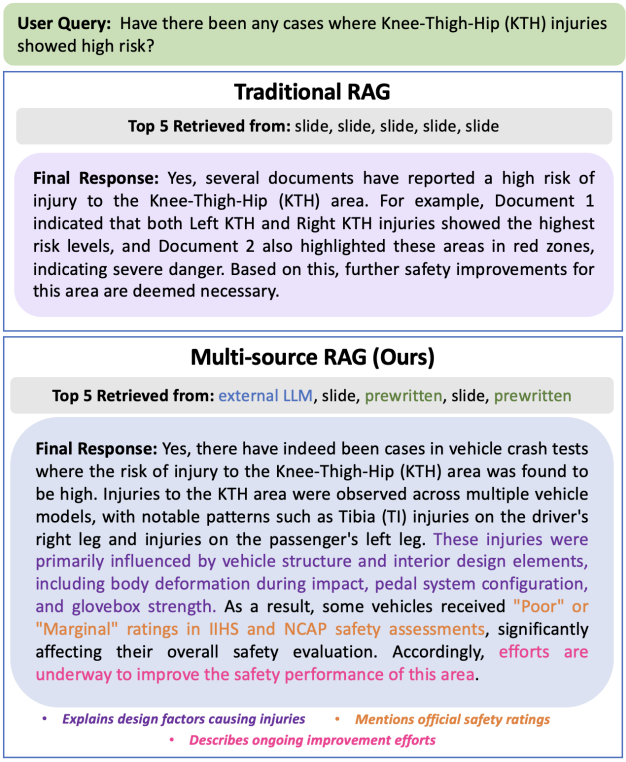


Figure A5. Comparison between Traditional RAG and SecMulti-RAG in QA task (translated from Korean).

References

1. Lewis, P.; Perez, E.; Piktus, A.; Petroni, F.; Karpukhin, V.; Goyal, N.; Küttler, H.; Lewis, M.; Yih, W.t.; Rocktäschel, T.; et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. In Proceedings of the 34th International Conference on Neural Information Processing Systems (NIPS '20), Vancouver, BC, Canada, 6–12 December 2020; Curran Associates Inc.: Red Hook, NY, USA, 2020.
2. OpenAI; Achiam, J.; Adler, S.; Agarwal, S.; Ahmad, L.; Akkaya, I.; Aleman, F.L.; Almeida, D.; Altenschmidt, J.; Altman, S.; et al. GPT-4 Technical Report. *arXiv* **2024**, arXiv:2303.08774.
3. Anthropic. The Claude 3 Model Family: Opus, Sonnet, Haiku. 2024. Available online: https://api.semanticscholar.org/CorpusID:268232499 (accessed on 5 January 2025).
4. DeepSeek-AI; Guo, D.; Yang, D.; Zhang, H.; Song, J.; Zhang, R.; Xu, R.; Zhu, Q.; Ma, S.; Wang, P.; et al. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. *arXiv* **2025**, arXiv:2501.12948.
5. Zhou, Y.; Liu, Z.; Dou, Z. AssistRAG: Boosting the Potential of Large Language Models with an Intelligent Information Assistant. *arXiv* **2024**, arXiv:2411.06805. [CrossRef]
6. Gutiérrez, B.J.; Shu, Y.; Gu, Y.; Yasunaga, M.; Su, Y. HippoRAG: Neurobiologically Inspired Long-Term Memory for Large Language Models. *arXiv* **2025**, arXiv:2405.14831.
7. Jeong, S.; Baek, J.; Cho, S.; Hwang, S.J.; Park, J.C. Adaptive-RAG: Learning to Adapt Retrieval-Augmented Large Language Models through Question Complexity. *arXiv* **2024**, arXiv:2403.14403.
8. Yu, W.; Iter, D.; Wang, S.; Xu, Y.; Ju, M.; Sanyal, S.; Zhu, C.; Zeng, M.; Jiang, M. Generate rather than Retrieve: Large Language Models are Strong Context Generators. *arXiv* **2023**, arXiv:2209.10063. [CrossRef]
9. Wu, R.; Chen, S.; Su, X.; Zhu, Y.; Liao, Y.; Wu, J. A Multi-Source Retrieval Question Answering Framework Based on RAG. *arXiv* **2024**, arXiv:2405.19207. [CrossRef]
10. Zhang, S.; Ye, L.; Yi, X.; Tang, J.; Shui, B.; Xing, H.; Liu, P.; Li, H. "Ghost of the past": Identifying and resolving privacy leakage from LLM's memory through proactive user interaction. *arXiv* **2024**, arXiv:2410.14931.
11. Kim, S.; Yun, S.; Lee, H.; Gubri, M.; Yoon, S.; Oh, S.J. ProPILE: Probing Privacy Leakage in Large Language Models. *arXiv* **2023**, arXiv:2307.01881. [CrossRef]
12. Hayes, J.; Melis, L.; Danezis, G.; Cristofaro, E.D. LOGAN: Evaluating Privacy Leakage of Generative Models Using Generative Adversarial Networks. *arXiv* **2017**, arXiv:1705.07663.
13. Lukas, N.; Salem, A.; Sim, R.; Tople, S.; Wutschitz, L.; Zanella-Béguelin, S. Analyzing Leakage of Personally Identifiable Information in Language Models. *arXiv* **2023**, arXiv:2302.00539. [CrossRef]

14. Chong, C.J.; Hou, C.; Yao, Z.; Talebi, S.M.S. Casper: Prompt Sanitization for Protecting User Privacy in Web-Based Large Language Models. *arXiv* **2024**, arXiv:2408.07004. [CrossRef]
15. Zhou, X.; Lu, Y.; Ma, R.; Gui, T.; Zhang, Q.; Huang, X. TextMixer: Mixing Multiple Inputs for Privacy-Preserving Inference. In Proceedings of the Findings of the Association for Computational Linguistics: EMNLP 2023, Singapore, 6–10 December 2023; Bouamor, H., Pino, J., Bali, K., Eds.; Association for Computational Linguistics: Stroudsburg, PA, USA, 2023; pp. 3749–3762. [CrossRef]
16. Yang, A.; Yang, B.; Zhang, B.; Hui, B.; Zheng, B.; Yu, B.; Li, C.; Liu, D.; Huang, F.; Wei, H.; et al. Qwen2. 5 Technical Report. *arXiv* **2024**, arXiv:2412.15115.
17. Douze, M.; Guzhva, A.; Deng, C.; Johnson, J.; Szilvasy, G.; Mazaré, P.E.; Lomeli, M.; Hosseini, L.; Jégou, H. The Faiss library. *arXiv* **2025**, arXiv:2401.08281.
18. Choi, N.; Byun, G.; Chung, A.; Paek, E.S.; Lee, S.; Choi, J.D. Reference-Aligned Retrieval-Augmented Question Answering over Heterogeneous Proprietary Documents. *arXiv* **2025**, arXiv:2502.19596.
19. Mizuno, K. *Crash Safety of Passenger Vehicles*; Translated from Japanese by Kyungwon Song; Reviewed by Inhwan Han, Jongjin Park, Sungjin Kim, Jongchan Park, and Namgyu Park; Bomyung Books: Seoul, Republic of Korea, 2016.
20. Chen, J.; Xiao, S.; Zhang, P.; Luo, K.; Lian, D.; Liu, Z. M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation. In Proceedings of the Findings of the Association for Computational Linguistics ACL 2024, Bangkok, Thailand and Virtual Meeting, 11–16 August 2024; Ku, L.W., Martins, A., Srikumar, V., Eds.; Association for Computational Linguistics: Stroudsburg, PA, USA, 2024; pp. 2318–2335. [CrossRef]
21. Xiao, S.; Liu, Z.; Zhang, P.; Xing, X. LM-Cocktail: Resilient Tuning of Language Models via Model Merging. *arXiv* **2023**, arXiv:2311.13534. [CrossRef]
22. Zhang, P.; Xiao, S.; Liu, Z.; Dou, Z.; Nie, J.Y. Retrieve Anything To Augment Large Language Models. *arXiv* **2023**, arXiv:2310.07554. [CrossRef]
23. Xiao, S.; Liu, Z.; Zhang, P.; Muennighoff, N. C-Pack: Packaged Resources To Advance General Chinese Embedding. *arXiv* **2023**, arXiv:2309.07597.
24. Zheng, L.; Chiang, W.L.; Sheng, Y.; Zhuang, S.; Wu, Z.; Zhuang, Y.; Lin, Z.; Li, Z.; Li, D.; Xing, E.P.; et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *arXiv* **2023**, arXiv:2306.05685.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.